WORKED SOLUTION

STEP I 2015, Question 13

Probability and Statistics • Conditional Probability

STEP I 2015, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q13

The key idea

Condition on when the first 6 appears. If we know the first 6 is on throw n, then each of the earlier $n-1$ throws is equally likely to be any of $1,\dots,5$, independently, so questions about 4s and 5s before the first 6 become ordinary binomial questions. Summing over n then gives geometric series and their derivatives, which is exactly where the binomial expansion in the hint comes in.

Solution

Throws are independent, each face having probability $\frac{1}{6}$. Write $p_n = \mathrm{P}(A)$ for the event that the first 6 is on throw n.

Part (i)

The first 6 is on throw n exactly when throws $1,\dots,n-1$ are all non-6 and throw n is a 6. By independence,

$$\mathrm{P}(A) = p_n = \left(\tfrac{5}{6}\right)^{n-1}\tfrac{1}{6}, \qquad n = 1,2,3,\dots$$

The conditioning fact. Given A, each of throws $1,\dots,n-1$ is known only to be "not a 6". Since the die is fair and throws are independent, these $n-1$ throws are independent and each is uniform on $\{1,2,3,4,5\}$. Every later part uses this, together with the law of total probability $\mathrm{P}(X) = \sum_n \mathrm{P}(X \mid A)\, p_n$.

Part (ii)

It is easier to find $\mathrm{P}(B')$, that no 5 comes before the first 6. Given A, each of the $n-1$ earlier throws avoids 5 with probability $\frac{4}{5}$, so

$$\mathrm{P}(B') = \sum_{n=1}^{\infty} \left(\tfrac{4}{5}\right)^{n-1} \left(\tfrac{5}{6}\right)^{n-1} \tfrac{1}{6} = \tfrac{1}{6} \sum_{n=1}^{\infty} \left(\tfrac{2}{3}\right)^{n-1} = \tfrac{1}{6} \cdot \frac{1}{1-\frac{2}{3}} = \tfrac{1}{2}.$$

Hence $\mathrm{P}(B) = \frac{1}{2}$.

Part (iii)

By the inclusion–exclusion principle (in complement form),

$$\mathrm{P}(B \cap C) = 1 - \mathrm{P}(B') - \mathrm{P}(C') + \mathrm{P}(B' \cap C').$$

By symmetry between the faces 4 and 5, $\mathrm{P}(C') = \mathrm{P}(B') = \frac{1}{2}$. For $B' \cap C'$, given A each earlier throw must avoid both 4 and 5, with probability $\frac{3}{5}$:

$$\mathrm{P}(B' \cap C') = \sum_{n=1}^{\infty} \left(\tfrac{3}{5}\right)^{n-1} \left(\tfrac{5}{6}\right)^{n-1} \tfrac{1}{6} = \tfrac{1}{6} \cdot \frac{1}{1-\frac{1}{2}} = \tfrac{1}{3}.$$

So $\mathrm{P}(B \cap C) = 1 - \frac{1}{2} - \frac{1}{2} + \frac{1}{3} = \frac{1}{3}$.

Part (iv)

Given A, the number of 5s among the $n-1$ earlier throws is $\mathrm{B}(n-1, \frac{1}{5})$, so

$$\mathrm{P}(D \mid A) = (n-1)\tfrac{1}{5}\left(\tfrac{4}{5}\right)^{n-2} \qquad (n \geq 2),$$

and it is 0 for $n = 1$. Hence

$$\mathrm{P}(D) = \sum_{n=2}^{\infty} (n-1)\tfrac{1}{5}\left(\tfrac{4}{5}\right)^{n-2} \left(\tfrac{5}{6}\right)^{n-1} \tfrac{1}{6}.$$

Pull out one factor of $\frac{5}{6}$ and pair the rest: $\frac{1}{5} \cdot \frac{5}{6} \cdot \frac{1}{6} = \frac{1}{36}$ and $\frac{4}{5} \cdot \frac{5}{6} = \frac{2}{3}$. With $m = n-1$,

$$\mathrm{P}(D) = \tfrac{1}{36} \sum_{m=1}^{\infty} m \left(\tfrac{2}{3}\right)^{m-1} = \tfrac{1}{36} \left(1 - \tfrac{2}{3}\right)^{-2} = \tfrac{9}{36} = \tfrac{1}{4},$$

using the given expansion with exponent -2: $(1-x)^{-2} = 1 + 2x + 3x^2 + \cdots$.

? **Ask yourself** How do we get $\mathrm{P}(D \cup E)$ without listing all the ways that one or both can happen?

Part (v)

Use $\mathrm{P}(D \cup E) = \mathrm{P}(D) + \mathrm{P}(E) - \mathrm{P}(D \cap E)$, with $\mathrm{P}(E) = \mathrm{P}(D) = \frac{1}{4}$ by the 4–5 symmetry. So we only need $\mathrm{P}(D \cap E)$.

Given A, exactly one 4 and exactly one 5 among the $n-1$ earlier throws means choosing which throw is the 5 and which is the 4 ($(n-1)(n-2)$ ordered choices), with the remaining $n-3$ throws all in $\{1,2,3\}$:

$$\mathrm{P}(D \cap E \mid A) = (n-1)(n-2)\left(\tfrac{1}{5}\right)^2\left(\tfrac{3}{5}\right)^{n-3} \qquad (n \geq 3).$$

Then

$$\mathrm{P}(D \cap E) = \sum_{n=3}^{\infty} (n-1)(n-2)\tfrac{1}{25}\left(\tfrac{3}{5}\right)^{n-3}\left(\tfrac{5}{6}\right)^{n-1}\tfrac{1}{6}.$$

Taking out $\left(\frac{5}{6}\right)^2$ gives the constant $\frac{1}{25}\cdot\frac{25}{36}\cdot\frac{1}{6} = \frac{1}{216}$, and $\frac{3}{5}\cdot\frac{5}{6} = \frac{1}{2}$. With $k = n-3$,

$$\begin{aligned}\mathrm{P}(D \cap E) &= \tfrac{1}{216}\sum_{k=0}^{\infty}(k+2)(k+1)\left(\tfrac{1}{2}\right)^k = \tfrac{2}{216}\sum_{k=0}^{\infty}\frac{(k+2)!}{k!\,2!}\left(\tfrac{1}{2}\right)^k \\ &= \tfrac{1}{108}\left(1-\tfrac{1}{2}\right)^{-3} = \tfrac{8}{108} = \tfrac{2}{27},\end{aligned}$$

using the given expansion with exponent -3, whose coefficient of x^k is $\frac{(k+2)!}{k!\,2!}$. Therefore

$$\mathrm{P}(D \cup E) = \tfrac{1}{4} + \tfrac{1}{4} - \tfrac{2}{27} = \tfrac{27-4}{54} = \tfrac{23}{54}.$$

Checks

There is a slicker way to see every answer, which makes an excellent check. Ignore all throws of 1, 2, 3 (and, for (ii) and (iv), also 4): they never affect the events. The remaining relevant throws form a sequence in which each relevant face is equally likely.

- (ii): the first throw that is a 5 or a 6 is equally likely to be either, so $\mathrm{P}(B) = \frac{1}{2}$.
- (iii): among 4, 5, 6, each of the three faces is equally likely to be the last of them to make its first appearance, and $B \cap C$ says 6 is last, so $\frac{1}{3}$.
- (iv): D means the 5-or-6 sequence starts 5, 6: probability $\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$.
- (v): $D \cap E$ means the 4-5-6 sequence starts $4,5,6$ or $5,4,6$: probability $2\left(\frac{1}{3}\right)^3 = \frac{2}{27}$.

All five answers were also confirmed by a computer simulation of 3×10^5 sequences of throws, which gave $\mathrm{P}(D \cup E) \approx 0.426$, against $\frac{23}{54} \approx 0.4259$.

Common slips

- Treating the earlier throws as uniform on all six faces: given A, they cannot be 6, so each 5 has probability $\frac{1}{5}$, not $\frac{1}{6}$.
- Forgetting the factor $(n-1)(n-2)$ in (v), or using $\binom{n-1}{2}$, which does not distinguish which of the two throws is the 4.
- Adding $\mathrm{P}(D) + \mathrm{P}(E)$ for (v) without subtracting the overlap.

Beyond STEP

Conditioning on a random stopping time and summing, as here, is the law of total probability in its most useful form, and the series you met are generating functions for the negative binomial distribution. The "ignore the irrelevant throws" check is an instance of a powerful idea: a process watched only at the moments something relevant happens is often far simpler than the original. Grimmett and Welsh, *Probability: An Introduction*, develops both ideas.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.